

$^{32}\text{S}(\alpha, p\gamma)$ 1970Ho09, 1972Br33, 1973Al22

$J^\pi=0^+$ for ^{32}S ground state.

1970Ho09: 10.975-, 11.960-, 12.120-MeV α beams were produced from the Chalk River MP tandem accelerator. Targets were 100- $\mu\text{g}/\text{cm}^2$ Sb_2S_3 on a 10- $\mu\text{g}/\text{cm}^2$ carbon backing. γ rays were detected using 15.2-cm-long by 12.7-cm-diam NaI(Tl) detectors. Protons were detected using an annular silicon surface barrier detector close to 180° with FWHM=30 keV for protons and FWHM=50 keV for α . Measured E_p , E_γ , I_γ , $p\gamma(\theta)$. Deduced 25 level energies from E_p , J, branching ratios, multipole mixing ratios.

1972Br33: a 10-MeV α beam was produced from the Chalk River MP tandem accelerator. Targets were 340- $\mu\text{g}/\text{cm}^2$ Sb_2S_3 on gold backings. γ rays were detected using a 44- cm^3 Ge(Li) detector inside an anticoincidence split annular NaI(Tl) detector. Measured E_γ , I_γ . Deduced 11 level energies, 16 γ branching ratios, 11 lifetimes using the Doppler Shift Attenuation Method (DSAM). **1970In01** from the same experiment deduced the lifetime using DSAM for a level at 3942 keV and the mixing ratio for a 3942 $\rightarrow$ 2646 γ transition.

1973Al22: a 14-MeV α beam was produced from the 80-cm cyclotron at the Research Institute for Physics in Stockholm and 12-16-MeV α beams were produced from the Uppsala tandem accelerator. Targets were prepared by depositing pulverized sulphur enriched to 99.9% in ^{32}S on lead backings. γ rays were detected using an Ortec 60- cm^3 Ge(Li) detector with FWHM=2.2 keV at 1333 keV and an additional Canberra 60- cm^3 Ge(Li) detector for $\gamma\gamma$ -coincidence measurements. Measured E_γ , I_γ , $\gamma\gamma$ -coin, $\gamma(\theta)$ for 4 γ rays. Deduced 12 level energies from E_γ , J for 4 levels, branching ratios, and mixing ratios.

1969In04: 6-8-MeV α beams were produced from the Chalk River MP tandem accelerator. A thick target (270 $\mu\text{g}/\text{cm}^2$) of natural sulfur on a tantalum backing and a thin target (50 $\mu\text{g}/\text{cm}^2$) of Sb_2S_3 on a gold backing were used. γ rays were detected using two Ge(Li) detectors of 15 and 40 cm^3 volume. Measured E_γ , I_γ . Deduced 6 level energies, branching ratios, 5 lifetimes using the Doppler Shift Attenuation Method (DSAM) and 1 lifetime for the 3163-keV level using the Recoil Distance Method (RDM), and transition strengths.

1974Lo17: 11.0, 12.0 and 14.7-MeV α beams were produced from the Oliver Lodge Laboratory at the University of Liverpool. Targets were natural CdS. Protons were detected using an annular ΔE -E telescope close to 180° with FWHM=80 keV. γ rays were detected using five NaI(Tl) detectors and one Ge(Li) detector. Measured E_γ , I_γ , $p\gamma$ -coin, $p\gamma(\theta)$, $\gamma\gamma$ -coin, γ -ray linear polarization. Deduced J and π for 3 levels, 6 γ branching ratios, 5 mixing ratios, and lifetimes for a 5408-keV level using DSAM and for a 6088-keV level using RDM.

1973Br26: a 11.2-MeV α beam was produced from the Legnaro (Padova) Van de Graaff generator. Targets were natural CdS. γ rays were detected using a Ge(Li) detector. Deduced the lifetime using the Recoil Distance Method (RDM) for a level at 3163 keV.

1973An01: 5.5- and 8.6-MeV α beams of 100-200 nA were produced from the Oxford Vertical Van de Graaff (Injector) or the Oxford EN Tandem. Targets were natural sulphur in AgS. γ rays were detected using a Ge(Li) detector at 0° . Deduced the lifetime using the Recoil Distance Method (RDM) for a level at 3163 keV.

 ^{35}Cl Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}$	Comments
0	$3/2^+$		J^π : from the Adopted Levels.
1219.4 6	$1/2^+$	0.15 ps +7-6	J^π : from the Adopted Levels. $T_{1/2}$: Lifetime $\tau=0.21$ ps +10-8 (1969In04 DSAM).
1763.3 6	$5/2^+$	0.38 ps 11	J^π : spin=5/2 from $p\gamma(\theta)$ in 1970Ho09 ; parity=+ from 1763.2 γ M1+E2 to $3/2^+$. $T_{1/2}$: Lifetime $\tau=0.55$ ps 15 from 1969In04 using DSAM.
2645.8 8	$7/2^+$	0.21 ps 6	J^π : spin=7/2 from $p\gamma(\theta)$ in 1970Ho09 ; parity from 882 γ , M1+E2 to $5/2^+$ in 1970Ho09 . $T_{1/2}$: Lifetime $\tau=0.30$ ps 9 from 1969In04 using DSAM.
2693.6 13	$3/2^+, 5/2^+$	49 fs 21	$T_{1/2}$: Lifetime $\tau=70$ fs 30 from 1972Br33 using DSAM. Other: $\tau < 30$ fs (1969In04 , DSAM).
3002.6 15	$(5/2)^+$	33 fs 6	J^π : spin=3/2, 5/2 from $p\gamma(\theta)$ in 1970Ho09 ; parity from 3002.5 γ , M1+E2 to $3/2^+$ in 1970Ho09 ; J=3/2 is disfavored since it would give a large $\delta(E2/M1)=1.2$ 5 for 3002.5 γ in 1970Ho09 which is less likely and also due to the absence of an expected γ to 1219.5, $1/2^+$ level if J=3/2. $T_{1/2}$: Lifetime $\tau=48$ fs 9 from 1972Br33 using DSAM. Other: $\tau < 50$ fs (1969In04 , DSAM).
3163.1 11	$7/2^-$	42 ps 5	J^π : spin=7/2 from $p\gamma(\theta)$ in 1970Ho09 ; parity from 3162.6 γ , M2+E3 to $3/2^+$ in 1970Ho09 .

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$^{32}\text{S}(\alpha, p\gamma)$ **1970Ho09, 1972Br33, 1973Al22 (continued)** ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2}	Comments
3915 2		<15.3 fs	T _{1/2} : Lifetime $\tau=60$ ps 7 from 1969In04. E(level): from 1972Br33.
3943.4 12	9/2	0.229 ps 35	T _{1/2} : Lifetime $\tau<22$ fs from 1972Br33 using DSAM.
3968 2		<37 fs	T _{1/2} : Lifetime $\tau=0.33$ ps 5 from 1972Br33 using DSAM. E(level): from 1972Br33.
4058 3		22 fs 9	T _{1/2} : Lifetime $\tau<54$ fs from 1972Br33 using DSAM.
4108 2	3/2, 7/2	49 fs 11	E(level): weighted average of 4045 10 (1970Ho09) and 4058 2 (1972Br33). T _{1/2} : Lifetime $\tau=31$ fs 13 from 1972Br33 using DSAM.
4171 2		47 fs 17	E(level): weighted average of 4110 10 (1970Ho09) and 4108 2 (1972Br33). T _{1/2} : Lifetime $\tau=70$ fs 16 from 1972Br33 using DSAM.
4347.8 12	9/2 ⁻	2.0 ps +10-5	E(level): weighted average of 4175 10 (1970Ho09) and 4171 2 (1972Br33). T _{1/2} : Lifetime $\tau=68$ fs 24 from 1972Br33 using DSAM.
4618 2	3/2, 5/2	40 fs 17	J ^π : spin from $p\gamma(\theta)$ in 1974Lo17; parity from 1184.6 γ , M1+E2 to 7/2 ⁻ in 1974Lo17. Other: J=5/2, 9/2 from $p\gamma(\theta)$ in 1970Ho09 and $p\gamma(\theta)$ in 1973Al22. T _{1/2} : Lifetime $\tau=2.9$ ps +14-7 from 1972Br33 using DSAM.
4770 [#] 15	7/2, 9/2		E(level): from 1972Br33.
4880.8 21	7/2	0.19 ps 4	T _{1/2} : Lifetime $\tau=58$ fs 24 from 1972Br33 using DSAM.
5015 [#] 20			E(level): other: 4882 2 from 1972Br33, 4885 20 from 1970Ho09.
5175 [#] 20			T _{1/2} : Lifetime $\tau=0.28$ ps 6 from 1972Br33 using DSAM.
5230 [#] 20	1/2, 3/2, 5/2		
5407.1 15	11/2 ⁻	0.28 ps 7	E(level): other: 5415 20 from 1970Ho09. J ^π : spin from $p\gamma(\theta)$ in 1974Lo17 and from $\gamma(\theta)$ and yield function in 1973Al22; parity from 1059.3 γ , M1+E2 to 9/2 ⁻ in 1974Lo17. T _{1/2} : Lifetime $\tau=0.4$ ps 1 from 1974Lo17 using DSAM.
5535 [#] 20			
5600 [#] 20			
5650 [#] 20	5/2, 9/2		
5850 [#] 25	5/2, 9/2		
5927.1 19	7/2, 11/2		E(level): other: 5930 25 from 1970Ho09. J ^π : spin from $\gamma(\theta)$ in 1973Al22.
6087.0 18	13/2 ⁻	6.5 ps 6	J ^π : spin from $p\gamma(\theta)$ in 1974Lo17; parity from 679.9 γ , M1+E2 to 11/2 ⁻ in 1974Lo17. Other: J=9/2, 13/2 from $\gamma(\theta)$ in 1973Al22. T _{1/2} : Lifetime $\tau=9.3$ ps 8 from 1974Lo17 using RDM.
6400 [#] 25			

[†] From a least-squares fit to E_γ values with uncertainties, unless otherwise noted.[‡] Spin assignments are from $p\gamma(\theta)$ in 1970Ho09, unless otherwise noted.[#] From 1970Ho09, derived from measured E_p. $\gamma(^{35}\text{Cl})$

E _i (level)	J ^π _i	E _γ [†]	I _γ [‡]	E _f	J ^π _f	Mult. [#]	δ [#]	Comments
1219.4	1/2 ⁺	1219.4 6	100	0	3/2 ⁺			E _γ : weighted average of 1219.3 6 (1969In04) and 1219.5 10 (1973Al22).
1763.3	5/2 ⁺	543.9	<0.5	1219.4	1/2 ⁺			
		1763.2 6	100	0	3/2 ⁺	M1+E2	3.0 1	E _γ : from 1969In04. Other: 1763.1 10 (1973Al22).
2645.8	7/2 ⁺	882.4 10	11.1 22	1763.3	5/2 ⁺	M1+E2	+0.25 5	I _γ : from 1970Ho09. Other: 18 (1969In04), 10 (1973Al22).

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$^{32}\text{S}(\alpha, p\gamma)$ **1970Ho09, 1972Br33, 1973Al22 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	Comments
2645.8	$7/2^+$	1426.4 2646.2 15	<1.2 100.0 22	1219.4 0	$1/2^+$ $3/2^+$			E_γ : from 1969In04. Other: 2646.0 20 (1973Al22). I_γ : from 1970Ho09. Other: 100 (1969In04), 100 (1973Al22). I_γ : from 1970Ho09. I_γ : from 1970Ho09. Other: 10 (1969In04). I_γ : from 1970Ho09. Other: 100 (1969In04), 100 (1973Al22). δ : $\delta=+0.17$ 8 for $J=3/2$; $\delta=-0.26$ 5 for $J=5/2$ (1970Ho09).
2693.6	$3/2^+, 5/2^+$	930.5 15 1474.2 2693.2 20	18.2 26 11.7 26 100 4	1763.3 1219.4 0	$5/2^+$ $1/2^+$ $3/2^+$	M1+E2		E_γ : from 1969In04. Other: 2646.0 20 (1973Al22). I_γ : from 1970Ho09. Other: 100 (1969In04), 100 (1973Al22). I_γ : from 1970Ho09. I_γ : from 1970Ho09. Other: 10 (1969In04). I_γ : from 1970Ho09. Other: 100 (1969In04), 100 (1973Al22). δ : $\delta=+0.17$ 8 for $J=3/2$; $\delta=-0.26$ 5 for $J=5/2$ (1970Ho09).
3002.6	$(5/2)^+$	309.0 356.8 1239.3 1783.2 3002.5 15	<5 <1 <2 <2 100	2693.6 2645.8 1763.3 1219.4 0	$3/2^+, 5/2^+$ $7/2^+$ $5/2^+$ $1/2^+$ $3/2^+$	M1+E2	+0.09 3	I_γ : other: <4 (1972Br33). I_γ : from 1969In04, 1972Br33. E_γ : weighted average of 3003 2 (1969In04), 3003 2 (1972Br33), and 3002.0 15 (1973Al22). δ : other: 1.2 5 if $J(3002.6)=3/2$ (1970Ho09).
3163.1	$7/2^-$	160.5 469.5 & 517.4 15 1399.8 3162.6 15	<0.54 <0.54 11.1 33 <1.1 100.0 33	3002.6 2693.6 2645.8 1763.3 0	$(5/2)^+$ $3/2^+, 5/2^+$ $7/2^+$ $5/2^+$ $3/2^+$	M2+E3	-0.14 2	I_γ : from 1970Ho09. Other: 8.7 (1969In04), 11 (1973Al22). I_γ : from 1970Ho09. Other: 100 (1969In04), 100 (1973Al22).
3915		2696 3915	19 5 100 5	1219.4 0	$1/2^+$ $3/2^+$			
3943.4	$9/2$	1297.4 15 2180.2 15	8.1 21 100.0 21	2645.8 1763.3	$7/2^+$ $5/2^+$			I_γ : weighted average of 9.9 22 (1970Ho09) and 6.4 21 (1972Br33). Other: 11 (1973Al22). I_γ : others: 100.0 22 (1970Ho09), 100 (1973Al22).
3968		2749 3968	100 <2	1219.4 0	$1/2^+$ $3/2^+$			
4058		2295 2839 4058	8 4 100 4 <2.2	1763.3 1219.4 0	$5/2^+$ $1/2^+$ $3/2^+$			
4108	$3/2, 7/2$	2345 4108	16 5 100 5	1763.3 0	$5/2^+$ $3/2^+$			
4347.8	$9/2^-$	1184.6 10 1702.1 15	100.0 31 52 7	3163.1 2645.8	$7/2^-$ $7/2^+$	M1+E2 E1(+M2)	-0.36 3 0.00 3	I_γ : from 1974Lo17. Others: 100 7 (1970Ho09) and 100 4 (1972Br33). δ : from 1974Lo17. Other: -0.40 +8-9 (1973Al22); +0.42 4 (1970Ho09). $A_2=-0.95$ 3, $A_4=+0.08$ 3 (1974Lo17). $\text{POL}=-0.005$ 40 (1974Lo17). I_γ : unweighted average of 61 7 (1970Ho09), 39 4 (1972Br33), and 56.3 31 (1974Lo17). Mult., δ : D(+Q) and δ from $p\gamma(\theta)$ and electric character from $\gamma(\text{pol})$ in 1974Lo17. $A_2=-0.34$ 5, $A_4=+0.04$ 5 (1974Lo17). $\text{POL}=+0.36$ 7 (1974Lo17).

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$^{32}\text{S}(\alpha, p\gamma)$ **1970Ho09, 1972Br33, 1973Al22 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	Comments
4618	3/2, 5/2	4618 2	100	0	3/2 ⁺			E_γ, I_γ : from 1972Br33. δ : +0.08 14 for J=3/2; -0.36 15 for J=5/2 (1970Ho09).
4880.8	7/2	2187.1 @	52 @ 9	2693.6	3/2 ⁺ , 5/2 ⁺			E_γ, I_γ : Final level is uncertain, either 2646 or 2694 (1970Ho09).
		2234.9 @	52 @ 9	2645.8	7/2 ⁺			E_γ, I_γ : Final level is uncertain, either 2646 or 2694 (1970Ho09).
		3117.4 20	100 9	1763.3	5/2 ⁺			I_γ : from 1970Ho09. Other: 100 (1973Al22).
5230	1/2, 3/2, 5/2	5230	100	0	3/2 ⁺			
5407.1	11/2 ⁻	1059.3 10	9.9 22	4347.8	9/2 ⁻	M1+E2	+0.25 8	Mult., δ : D+Q and δ from $\gamma(\theta)$ and magnetic character from $\gamma(\text{pol})$ (1974Lo17). $A_2=+0.19$ 11, $A_4=-0.01$ 13 (1974Lo17). POL=-0.40 14 (1974Lo17). Mult.: Q(+O) with $\delta(\text{O/Q})=0.00$ 3 from $\text{py}(\theta)$ in 1974Lo17; M2 ruled out by RUL. Other: Q(+O) with $\delta(\text{O/Q})=+0.02$ 9 (1973Al22). $A_2=+0.44$ 3, $A_4=-0.19$ 3 (1974Lo17).
		2244 3	100.0 22	3163.1	7/2 ⁻	E2		
5850	5/2, 9/2	2687	100	3163.1	7/2 ⁻	D+Q		Mult., δ : from $\text{py}(\theta)$ in 1970Ho09 with $\delta(\text{Q/D})=-2.2$ 5 or -0.6 1 for J=5/2; +0.28 4 for J=9/2 (1970Ho09).
5927.1	7/2, 11/2	1579.3 15	100	4347.8	9/2 ⁻	D+Q		Mult., δ : from $\gamma(\theta)$ in 1973Al22, with $\delta(\text{Q/D})=+1.1$ 5 for J=7/2; -0.8 4 for J=11/2.
6087.0	13/2 ⁻	679.9 10	100 6	5407.1	11/2 ⁻	M1+E2	+0.020 15	Mult., δ : D+Q and δ from $\text{py}(\theta)$ and magnetic character from $\gamma(\text{pol})$ in 1974Lo17. Other: $\delta=0.00$ 7 for $J_i=13/2$ (1973Al22). $A_2=-0.22$ 1, $A_4=0.005$ 15 (1974Lo17). POL=-0.45 5 (1974Lo17).
		1739.2	18 6	4347.8	9/2 ⁻			

[†] Values with uncertainties are from 1973Al22 and those without uncertainties are from level-energy differences, unless otherwise noted. 1973Al22 is the only literature in this dataset that reported measured E_γ with uncertainties. 1969In04 and 1972Br33 reported $E(\text{level})$ with uncertainties derived from their measured E_γ but did not report E_γ . Evaluators adopt the uncertainties of $E(\text{level})$ from 1969In04 and 1972Br33 as the uncertainties of an E_γ if this γ transition populates the ground state and is also the dominant decay branch of that level.

[‡] From 1969In04 for γ rays from level up to 3163, from 1972Br33 for γ rays from level above 3163 up to 4618 level, and from 1974Lo17 for γ rays from level above 4618, unless otherwise noted.

[#] Mult=D+Q, Q or Q+O and δ are determined from $\text{py}(\theta)$ in 1970Ho09; magnetic or electric characters are determined based on RUL and measured level $T_{1/2}$ where available, unless otherwise noted.

@ Multiply placed with undivided intensity.

& Placement of transition in the level scheme is uncertain.

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Legend

Level Scheme

Intensities: Relative photon branching from each level
& Multiply placed: undivided intensity given

-----► γ Decay (Uncertain)